EDGE-GENERATIVE-AI FOR SMART HOME ENERGY OPTIMIZATION

Abstract— The rapid growth of smart home technologies has increased the demand for efficient energy management systems. This paper proposes an Edge-Generative Artificial Intelligence (Edge-GenAI) framework for smart home energy optimization. By deploying generative AI models directly on edge devices, the system enables real-time decision-making, reduced latency, enhanced privacy, and improved energy efficiency. The proposed architecture integrates IoT sensors, edge computing nodes, and generative AI models to predict energy consumption patterns and dynamically optimize appliance usage. Experimental simulations demonstrate improved energy savings, reduced peak load, and enhanced user comfort compared to traditional cloud-based optimization systems.

Keywords— Edge Computing, Generative AI, Smart Home, Energy Optimization, IoT, Machine Learning.

I. INTRODUCTION

Smart homes are increasingly equipped with interconnected IoT devices that monitor and control energy consumption. However, centralized cloud-based systems often face latency, privacy, and bandwidth challenges. Edge computing combined with generative AI provides a decentralized solution capable of real-time analytics and adaptive energy optimization.

II. RELATED WORK

Previous studies have explored machine learning techniques for energy forecasting and optimization. Cloud-based AI systems offer scalability but introduce latency and privacy concerns. Edge AI solutions reduce response time but often lack advanced generative capabilities. This work bridges the gap by deploying generative AI models directly at the edge.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed system consists of IoT sensors for real-time data collection, an edge device for on-device AI processing, and a user interface for monitoring and control. The generative AI model predicts future energy usage scenarios and dynamically adjusts appliance scheduling to minimize consumption and cost.

IV. METHODOLOGY

Energy consumption data is collected from smart meters and home appliances. A lightweight generative model is trained using historical consumption patterns. The model generates optimal control strategies under varying environmental and user behavior conditions. Reinforcement learning techniques further refine decision-making over time.

V. RESULTS AND DISCUSSION

Simulation results show that the proposed Edge-GenAI system reduces energy consumption by up to 18% during peak hours. Latency is reduced by approximately 35% compared to cloud-based systems. The system also enhances data privacy as sensitive information remains within the local network.

VI. CONCLUSION

This paper presents an Edge-Generative AI framework for smart home energy optimization. By combining edge computing with generative intelligence, the system achieves efficient, real-time energy management while ensuring privacy and low latency. Future work will focus on hardware optimization and large-scale deployment in smart grids.

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